

Seminars 1 & 2

Studying how protests grow (or decline)

-> Find factors (of what could make it grow or decline)

-> Agent-based model?

online vs physical protest (what drives them)?

Feedback:

- contagion in crowd
- left wings vs right wing protest
 - Based on Louis Theroux manosphere documentary.
 - The successfulness in gathering people.
 - define what a successful protest is
- find paper and base our model on it (be careful with a large language model)
- scale & scope
 - think of cultural differences
 - Based the protest on one country or one event
- intensity $\neq$ amount of protester
 - proportion of protestors opposed to the population
- activism $\neq$ protest (for online protest)
- slacktivism
- try to come up with concrete hypotheses.
 - later try to think of which/how model to use

1 - Problematization:

Why is our issue complex?

explain

draw CLDs

explain each variable

2 - Clear research question

3- Clear hypothesis

4 - Model

Literature

[Lemos, Carlos, Helder Coelho, and Rui J. Lopes. "Agent-based modeling of protests and violent confrontation: a micro-situational, multi-player, contextual rule-based approach." *Proceedings of the 5th World Congress on Social Simulation, São Paulo, Brazil. 2014.*](#)

Summary:

They created an ABM model of street protests with the actors being police officers and protesters with three personality types: ("hardcore, "hangers-on", and passers-by"). "Media" agents are there to publicise violent events.

- Agents have multiple goals
Action selection happens through a "personality" vector with context rules for adaptation
- Protesters turn active or violent according to a threshold rule proposed by Joshua Epstein's 2002 threshold rule.
 - Threshold rule: "a decision-making mechanism that determines whether an agent becomes active in a rebellion based on local conditions. The core of this rule is a threshold-based activation mechanism, often defined as a "Quiescent" (inactive) agent becoming "Active" if their personal grievance exceeds their perceived net risk from authorities".
- Police agents arrest violent protestors within their move range if they have enough police around them (something that has to be defined based on literature for us, depending on what types of protests or where we want to model them).

The model was applied to an event where policemen defended a government building from protesters. Several emergent crowd patterns prevalent in real protests appeared, such as clustering of violent and active protesters in the front (confrontation line) which moved back and forth with localised fights. Violent behaviour first occurred within the more aggressive protesters and did not propagate to the rest of the crowd (something we can possibly confirm!)

[Msimanga, H. H., and B. Mashwama. "Modeling and Simulating Student Protests Through Agent-Based Framework." *International Journal of Information Communication Technologies and Human Development*, vol. 15, no. 1, 2023.](#)

Summary:

This paper presents an agent-based model to study the effect of grievance, net risk, social, sympathy, and political influence on the likelihood of student protests emerging in South Africa universities. Studies of student protests in several fields have been conducted, but no ABM has been used to explore factors contributing to student protests. Student protests have proved to be disorderly, frequently leading to property damage, academic program cancellations, and injuries. Simulation experiments demonstrated that inequality level, number of activists, activist's influential size, number of friendship ties, suspend delay, and sympathy are elements that determine the model of social conflicts, since there are statistically significant in the logistic regression. For university administration to effectively handle disruptive student protest actions, risk management policies should focus on understanding network structures that integrate students' interactions to monitor the spread of opinions that initiate protest mobilization. They used netlogo to model this which we used as well.

GRAY-HAWKINS, M. (2018). COLLECTIVE MOVEMENTS, DIGITAL ACTIVISM, AND PROTEST EVENTS: THE EFFECTIVENESS OF SOCIAL MEDIA CONCERNING THE ORGANIZATION OF LARGE-SCALE POLITICAL PARTICIPATION. *Geopolitics, History, and International Relations*, 10(2), 64. <https://doi.org/10.22381/ghir10220188>

-> Social media doesn't just "inform" people — it actively shapes protest growth through networks, identity, coordination, and sometimes suppression.

Zhu, Q., M. Skoric, and F. Shen (2017). "I Shield Myself from Thee: Selective Avoidance on Social Media during Political Protests," *Political Communication* 34(1): 112-131.

-> protests are not random — they follow patterns of diffusion, growth, and decline over time, shaped by individual behavior and system-level dynamics.

<https://tmap.org/submission/index.php/tpm/article/view/641>

<https://www.sciencedirect.com/science/article/pii/S0305750X2100276X>

Summary:

The literature considers nonviolent protests among the most important predictors of transitions towards democracy and democratic reforms. This study addresses the conditions making countries more likely to experience nonviolent instead of violent forms of protest. While the literature emphasizes economic and political predictors of protest at the country level, we expand the study of nonviolent-vs-violent protest by incorporating cultural predictors. To do so, we use a newly developed time-pooled cross-sectional database covering an established set of orientations from the World Values Survey, known as "emancipative values". Estimating the prevalence of these values at the country level as a time-varying predictor of nonviolent-vs-violent protests, we present the first comprehensively longitudinal analysis of the determinants of protests. Taking into account time-varying unobserved heterogeneity, persistence, the excess number of zeros and over-dispersion in protest event data, we find that emancipative values are a significant and positive predictor of the countries' nonviolent protest levels, yet not violent protest levels.

Alsulami, A., Glukhov, A., Shishlenin, M., & Petrovskii, S. (2022). Dynamical modelling of street protests using the Yellow Vest Movement and Khabarovsk as case studies. *Scientific Reports*, 12(1), Article 20447. <https://doi.org/10.1038/s41598-022-23917-z>

- very similar cases

- Yellow Vest Protests done in response to increased fuel tax implementation from the government
 - 300,000 people protested
 - repeated every saturday for a year
 - disappeared end of 2019
- street protests in Khabarovsk in East Russia
 - russian government arrested regional politicians after a criminal investigation, but the regional leader was very popular
 - high peak and the gradual decrease in support of protests
- similar pattern in both protests :
 - Very large turnout at the start
 - Gradual decline over time
 - Eventual disappearance
- differences between two cases
 - the french case has a trough because of the christmas NYE period
- central question
 - Can a single general model explain protest dynamics across different societies?
 - author says yes
 - uses maths to figure out what factors matter most for protest intensity and duration

methods and models

- methods of geostatistics and regression analysis of the participants social traits
- describe the dynamics of protests by regarding it, as the dynamics of human decision-making and behaviour.
- are epidemics
- individuals 'catch' ideas and join the protests
- protesters are 'infected' when they get the idea to join a protest from a friend who is already protester
- or they get it from social media
- study the dynamics or evolution of opinion, fashion or habits of the protests

(Fig. 2): S are potential protest participants, I are the new protesters who joined the movement recently, C are the experienced (mature) protesters and R are the former protesters who became disappointed or disillusioned and withdrew from the movement. Note that the explicit inclusion of experienced protesters into the consideration as a separate group is an essential new feature of our model. The existence of such 'professional' protesters is admitted in the literature as an inherent part of the movement society (e.g. see³⁷⁻³⁹) but their actual role in the protests dynamics remains under-investigated.

Correspondingly, the model consists of the following equations:

$$\frac{dS}{dt} = -S(\beta_1 I + \beta_2 C) + \gamma R, \quad (1)$$

$$\frac{dI}{dt} = S(\beta_1 I + \beta_2 C) - \chi I - \delta_1 I, \quad (2)$$

$$\frac{dC}{dt} = \chi I - C\omega(I, C), \quad (3)$$

$$\frac{dR}{dt} = \delta_1 I + C\omega(I, C) - \gamma R. \quad (4)$$

The number of street protesters at time t is therefore $N(t) = I(t) + C(t)$. Each term in the right-hand side of Eqs. (1)–(4) describe the transition between the groups, the coefficients having the meaning of the average

- similar model of the SEIR epidemic model
- in a collective action group people learn from their experience and shift their behaviour to make themselves stronger
- social contagion (epidemic theory) is the more idea
- decision to protest spreads socially through:
 - Face-to-face interactions
 - Shared emotions
 - Social networks
 - Media and online communication
- this is the mechanism of diffusion

how people move between groups

- recruitment into a protest group
 - mature protesters more successful in recruiting
- new protesters become more mature protesters which makes them more radical
- people withdraw from the protest and become retired protesters
- people re-enter protests from 'retirement'
 - this creates a dynamic cycle rather than a one-time event

mature protesters:

- share a group identity
- share emotions
- loyalty
- social ties
- collective resistance

the larger the protest the less likely committed protesters are to leave

When crowds are big, morale rises and solidarity strengthens

Several parameters determine protest trajectories.

- β_1 = Recruitment by New Protesters
- β_2 = Recruitment by Mature Protesters
- χ = Rate New Protesters Become Mature
- δ_1 = Exit Rate of New Protesters
- δ_2 = Exit Rate of Mature Protesters
- γ = Re-entry Rate of Retired Protesters
- C_0 = Threshold Size

The protest size at which collective solidarity becomes important

n = Behavioural Threshold Sharpness

It determines how suddenly collective effects activate.

High n = people respond sharply once protest size passes a threshold.

Low n = response is gradual.

the model in the article is able to generate

- rapid collapse
- long slow decline
- stabilization
- growth after initial phase

bistability

- the same society can have a low protest equilibrium and a high protest equilibrium
 - depends on initial conditions
 - explains why similar grievances can lead to either a huge protest and movement or a minor 1 time protest

ghost attractor

- when movement is long and the participation slows only very slowly
- the decline accelerated unexpectedly

findings

- Khabarovsk participants appear to respond more strongly to collective thresholds
 - once enough people joined commitment sharply increased

- visible crowd size was very important for recruitment
- For Yellow Vests, the model estimates around 750,000 people eventually passed through the movement and withdrew.
 - protest influence was much broader than weekly attendance numbers alone suggest.
- protest endurance depends strongly on whether new joiners quickly lose motivation or become committed activists.

Vasist, P. N., Krishnan, S., Azad, N., & Mekoth, N. (2025). A Comparison of 'Organizing Forces' on Reddit and YouTube Networks: Insights From the France Riots of 2023. *Information Systems Frontiers*, 27(5), 2103–2118. <https://doi.org/10.1007/s10796-025-10587-1>

Beidollahkhani, A. (2025). Visual resistance, digital counterpublics, and feminist agency in Iran's digital protests from 2018 to 2023. *Discourse & Communication*, 19(5), 864–886. <https://doi.org/10.1177/17504813251380447>

<https://compass.onlinelibrary.wiley.com/doi/full/10.1111/soc4.12971>

Monshipouri, M., & Zamiri, R. (2023). Woman, Life, Freedom, One Year Later: Will the Iran Protests Succeed? *Middle East Policy*, 30(4), 10–25. <https://doi.org/10.1111/mepo.12722>

grievances were:

- economic crisis
- corruption
- repression
- lack of freedoms

- growing distance between the government and ordinary people

these grievances were despite the uproar after girl not wearing hijab was killed .

One of the biggest focuses of the article is Generation Z. Young Iranians, especially those born after the 1979 revolution, are described as the main force behind the protests.

They grew up in a digital world, are connected through social media, and compare their lives with people abroad. Because of that, they are less willing to accept strict controls and empty political promises.

they wanted real change

Women are at the heart of the movement, and this is what makes the protests so historically important.

Refusing compulsory hijab turned into a symbol of rejecting state control more broadly. The slogan Woman, Life, Freedom linked several ideas together:

- gender equality
- dignity
- freedom
- justice

The article also explains why these protests looked different from older ones. They were decentralized, meaning resistance was not only about huge demonstrations in Tehran. Protest and defiance happened in many spaces.

Examples included girls removing headscarves in classrooms, students protesting on campuses, and women continuing to walk unveiled in public even after mass demonstrations slowed down. This matters because it shows the movement continued through daily resistance, not just street crowds.

the state had:

- security forces
- arrests and repression
- surveillance technology
- internet shutdowns
- willingness to use violence

They also warn that even if the clerical regime weakened, democracy would not automatically appear.

Another important concept is the “gray class.” This refers to older or more cautious Iranians who may dislike the regime but do not actively protest because they fear instability, repression, or chaos. The authors argue that major transformation becomes much more possible if this silent majority joins the younger protesters.

In the end, the article says the movement's greatest strength is civil disobedience, the long-term refusal to obey unjust rules. Even without immediate regime change, the protests already changed Iranian politics by:

- breaking fear
- exposing the regime's legitimacy crisis
- empowering women
- showing youth rejection of the status quo
- proving a new future is imaginable

So even if the revolution did not "win" quickly, it may still have huge long-term importance.

After the protests; France's pension reform. (2010, October 30). *The Economist* (London), 397(8706), 55. -

https://lib.uva.nl/permalink/31UKB_UAM1_INST/1I0p012/cdi_gale_incontextgauss_KPI_A240721134

– more factual

Snow, D. A., & Moss, D. M. (2014). Protest on the Fly: Toward a Theory of Spontaneity in the Dynamics of Protest and Social Movements: Toward a Theory of Spontaneity in the Dynamics of Protest and Social Movements. *American Sociological Review*, 79(6), 1122-1143.

<https://www.tandfonline.com/doi/abs/10.1080/13510347.2016.1223630%4010.1080/tfocoll.2022.0.issue-best-paper-frank-cass-prize>

RQ + Work division

RQ

How do state responses to protest differ between the French pension protest in 2023 and the 2022 Iranian protests, and how do these responses affect protest demobilization?

H1: We expect the more oppression, the quicker the protests will spread

H2: State repression is more likely to lead to demobilization in authoritarian regimes than in democratic regimes.

H3: Repression perceived as unjust or excessive increases protest escalation through backlash effects.

For the final paper:

Task	Name	Deadline
Abstract	... ▾	06/05
Introduction	Julie ▾	06/05
Theory	Sophie ▾	06/05
Method	Maliah ▾	06/05
Results	Maliah ▾	06/05
Discussion	... ▾	06/05
Conclusion	... ▾	06/05

Matrix article

Paper Name	Main variable	Findings	Limitations
Brockett, C. D. (1993). A Protest-Cycle Resolution of the Repression/Popular-Protest Paradox. <i>Social Science History</i>, 17(3), 457–484. https://doi-org.proxy.uba.uva.nl/10.2307/1171433	IV: repression DV: mobilization Influence protest: - Emotions - Perceived injustice - Costs vs benefits of participation	Repression and protest relationship are not linear (repression can enhance or slow protest) Repression enhance protest when during mobilization phase (solidarity, anger, trigger participation) Repression can lead to protest never existing in a first place High repression can lead to radicalization (not just demobilization)	Based on central America Explain the problem at a macro level, but lack a micro explanation (individual decision-making) Rely heavily on the cycle model, when protests don't follow specific rules all the time
Schulte, F., & Steinert, C. V. (2023). Repression and backlash protests: Why leader arrests backfire. <i>International Interactions</i>, 49(1), 1–30. https://doi-org.proxy.uba.uva.nl/10.1080/03050629.2023.2149513	Iv: target of repression (leader arrests vs. mass arrests) Dv: occurrence of backlash protests (= a strong, often sudden, public reaction of opposition against a political, social, or cultural change)	Leader arrests can have backfire effect Not all repression has the same effect: <u>Leader arrests</u>: → more likely to increase protest <u>Mass arrests (ordinary participants)</u>: → no significant mobilizing effect → can even discourage participation Backlash depend on visibility and symbolism. When the state represses people (e.g. arrests a protest leader), it can unintentionally cause more people	Data limitations Only examine arrest (as a form of repression) To macro level oriented Shows correlation but not enough causality Does not take into account context variations

		to protest instead of scaring them away. ⚠ Backlash don't always occur so Repression ≠ automatic mobilization	
Pierre Rouxel, Karel Yon. From Pension Reform to Regime Crisis: The Social Movement of 2023 in France. Global Labour Journal, 2023, 14 (2), <10.15173/glj.v14i2.5519>. (hal-04385604	Article on the social movement in France 2023. Including how state response affects protesting.	Once institutional and legal constraints tighten and moderate union strategies hit a ceiling, mobilisation tends to shift from organised mass strikes to more autonomous/radical/ youth-driven street actions. Conclusion: agents should be able to change tactics and organisation.	I know this is not a research article, but I thought it could be helpful for background info.
Petrovskii S, Shishlenin M, Glukhov A (2025) Understanding street protests: from a mathematical model to protest management. PLoS ONE 20(4): e0319837. https://doi.org/10.1371/journal.pone.0319837	IV: police efficiency DV: intensity/duration (protesters are divided into groups of experience, necessary because they are likely to act differently) (gilets jaunes movement)	Increase in intensity of single police actions slightly decreased the number of daily protesters. However this only applies to novice protestors, not to experienced ones!	Protests and riots were not distinguished. Face to face communication whether to join the movement. Reliability of the data: number of protestors.

Model Brainstorming

Things we need to decide!

Define Scope

Temporal

- What is our idea of decline?
 - day-to-day changes?
 - pre/during/post event aggregate changes=

Geographic

- We want to do one city during one specific event, right?

Population

- Who are the agents?
- specific types of protestors, police, etc.
- divided by groups of experience (new and experienced)
- experienced were not leaving the protest as easily when police intervention happened (could be a rule)
-

Decline

- What counts as a decline for us? (find a paper for %)
 - A certain decrease in participants?
 - Shift to online protests?
 - Less protesting from that group?

Model Suitability

- ABM is better for modelling individuals who have decision rules

Steps to Take

- Make specific hypotheses for each agent type
- Ex: More police suppression brings protests online etc.
- Model one case first (probably the easier one)
- Create a bounded geography area (city district with a certain population)
- Figure out the time boundary (choose based on literature)
- Outcome: Proportion of people still participating over time
- We can simulate violent suppression vs. peaceful suppression of a protest
- Could we do this with NetLogo or Python?
- Change the parameters for the other case
- Run both at the same time
- See which parameters explain differing outcomes ex. (how the police act, how the protestors act, how dense the population is, etc.